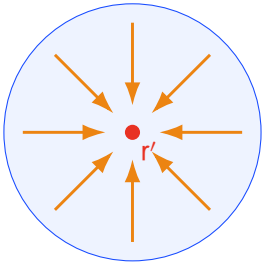
自由空间格林函数归一化：小球通量固定三维点源的系数与符号



半径 ϵ 的小球

$$G_0 = 1/(4\pi R), \qquad \nabla'^2 G_0 = 0 \quad (R > 0)$$

$$\oint \nabla' G_0 \cdot d\mathbf{S}' = -1, \text{ 故 } \nabla'^2 G_0 = -\delta^{(3)}(\mathbf{r} - \mathbf{r}')。$$

$$\rho = q\delta^{(3)}(\mathbf{r}' - \mathbf{r}_0) \Rightarrow \varphi(\mathbf{r}) = \frac{q}{4\pi\varepsilon_0 |\mathbf{r} - \mathbf{r}_0|}$$